

Golden Acres Marketplace Features and System Structure

This document serves as the technical blueprint for the Marketplace platform, outlining the essential features and priorities for our platform. It is a guide for development and assurance of the platform is robust, scalable, and built for the specific challenges of perishable logistics in Ghana.

1: Logistics Foundation (Non-Negotiable)

The highest priority is the technical integration that guarantees the integrity of our platform and the reliability of delivery. This is our core competitive advantage.

1. API Integration with 3PL (Driver Platform)

Goal: Fully automate the order-to-delivery handoff, eliminating manual data entry.

Requirements:

Real-Time Order Push: Seamless API endpoint to send order details (including GhanaPostGPS code, total weight, and "Refrigeration Required" flag) to the 3PL's dispatch system upon customer payment.

Tracking Number Webhook: The 3PL must immediately return a tracking number and estimated delivery time via a webhook back to our Order Management System (OMS) and customer account.

Status Sync: Continuous, real-time status updates (e.g., "Out for Delivery," "Delivered") to update the customer and our internal CS portal.

Proof of Delivery POD Capture: API must transfer the final geo-tagged Proof of Delivery (POD) photo and/or signature back to our system for audit trails.

2. Perishable Inventory Management (IMS)

Goal: Prevent overselling and manage the short shelf life of products.

Requirements:

Variable-Weight Pricing: The system must support listing products with an estimated weight and price range (e.g., yam, meat). The price should be automatically updated and reconciled post-picking but pre-payment capture (or credit/debit adjustment).

Live Farmer Sync: A simple, high-speed interface for farmers to update their exact available inventory in real-time. Failure to sync triggers low-stock alerts and eventually de-lists the item (enforcing the Farmer SOPs).

First Expiry, First Out (FEFO) Logic: Internal tracking logic that prioritizes the oldest batch of perishables for fulfillment to minimize waste at the aggregation hub.

3: Customer Experience & Retention

Features that build trust, drive high Average Order Value (AOV), and convert first-time buyers into loyal subscribers.

Checkout and Payment Gateway

Goal: Minimize friction and accommodate local payment preferences.

Requirements for Tech Team:

Multiple Local Payment Options: Must integrate with key Ghanaian Mobile Money (MoMo) providers (e.g., MTN, Vodafone) alongside standard Visa/Mastercard payment gateways.

Scheduled Delivery Picker: A calendar interface at checkout that displays available delivery time slots based on the 3PL's capacity and the product's availability (e.g., Saturday 8 AM–1 PM).

GhanaPostGPS Address Field: Mandatory, validated field for the GhanaPostGPS digital address code to ensure delivery accuracy.

4. Product Catalog & Merchandising

Goal: Make the shopping experience educational and inspiring.

Requirements:

Curated Bundles/Subscription Logic: Functionality to create and manage recurring Subscription Boxes ("Weekly Staples," "Recipe Kits") with automated recurring billing and order generation.

Farmer/Product Storytelling: Dedicated, editable profile pages for each farmer, including photos and growing methods, to build the trust and authenticity required in the food sector.

Side Recommendation

Related Content Widget: Recommendations for Ghanaian recipes using the items in the customer's cart.

5. Customer Service Portal (CS Portal)

Goal: Empower CS agents to resolve logistics and quality issues in under 30 minutes.

Requirements:

Unified Order Dashboard: Single-screen view showing Order Details, Payment Status, 3PL Tracking Log, and Driver ID.

Instant Refund Tool: A backend tool allowing CS agents to calculate and issue partial or full refunds/credits instantly for spoiled or missing items, linking the reason to the responsible party (Farmer, 3PL, or Hub etc).

6: Farmer Portal & Business Intelligence

Features that empower our farmers, drive retention, and provide data for the marketplace growth planning.

Farmer Management Portal

Goal: Provide farmers with a simple, transparent tool for inventory and payment.

Requirements for Tech Team:

Mobile-Optimized Interface: The portal must be fully functional and easy to use on a basic smartphone, as many farmers will not use a desktop computer.

Payment & Ledger View: Clear, accessible ledger showing total sales, commission deductions, applied penalties (SOP failures), and guaranteed net payout with the timestamp of the 48-hour payment.

Simple Product Uploader: Tool for listing products, uploading images (optimized for low bandwidth), and setting price/weight ranges.

7. Business Intelligence (BI) Reporting

Goal: Provide leadership with actionable data to optimize operations.

Requirements for Tech Team:

Core KPI Dashboards: Real-time visualization of: Customer Acquisition Cost (CAC) vs. Customer Lifetime Value (CLV), On-Time Delivery Rate (SLA Compliance), and Spoilage Rate by Farmer etc.

Demand Forecasting: Simple reports based on historical sales and seasonal trends to provide valuable planting and stock advice to our retained farmers.

8 Location Recommendation System: The 'MarketPlace Match'

I. System Goal: Building Density & Trust

The system will match users based on a calculated Proximity Score that prioritizes both distance (for logistics efficiency) and product availability (for customer satisfaction).

For Customers: Displaying farmers closest to their delivery address (using the GhanaPostGPS code, GPS) who have the items they are searching for.

For Farmers: Highlighting the most valuable delivery zones (clusters of high-demand customers) near their farm gate or aggregation hub.

II. Farmer Data Mapping (The Supply Side)

Farmers need to be mapped accurately to ensure logistical feasibility.

Feature

Primary Location Tag -GhanaPostGPS Address Code of the farm gate or their main aggregation/packing shed. Sets the fixed pickup point for the 3PL driver.

Secondary Location Tag -Identification of the closest urban market/town (e.g., "Kasoa Market") for a familiar anchor point. Used for marketing copy ("Freshly picked near [Town Name]...").

Delivery Radius Filter - Farmer - sets their acceptable "Farm-to-Hub" Radius (e.g., 50km). Used internally to filter out farmers too far from our main aggregation center, especially for bulk orders.

III. Customer Location Features (The Demand Side)

We use the compulsory GhanaPostGPS input to perform the matching.

Features

Pilot Zone Geo-Fence -When a customer inputs an address outside the pre-approved pilot zone, they are given a polite message: "We're not quite there yet but join the waitlist!" | Geo-fencing logic built into the checkout process linked to the 3PL's service map.

"Shop Local" Filter -An optional filter that defaults to showing products only from the 5 closest participating farmers to the customer's GhanaPostGPS code. Proximity Algorithm calculates distance between customer and all active farmer tags.

Delivery Fee Calculation -The delivery fee is dynamically calculated based on the distance between the GoldenAcres Aggregation Hub and the customer's GhanaPostGPS address. |API integration with a distance mapping tool (e.g., Google Maps API) to provide real-time distance to the 3PL.

9 Strategic and Marketing Implementation

Farmer Recommendation Marketing

We turn location matching into a powerful marketing tool:

- * Website Banner: "Your Yam is from Auntie Ama's farm, just 45km from your home!"
- * Filter Naming: Instead of just "All Produce," offer filters like "Shop Closest to East Legon" (for farmers located near the pilot area) and "Certified Organic Farms."

The core challenge for a virtual farmers market is Logistics Density (grouping many deliveries close together) and Trust. By connecting customers to their nearest participating farmers, we build that trust and reduce delivery costs.